R119 foundations: independent author proof draft

Internal mathematical validation copy

This standalone draft checks the proposed module. The integrated Econometrica manuscript supplies the retained primitive formulas, compiler, and complete Brownian Picard proof referred to here.

Well-posed original decisions. The comparison uses a specified conditional nested recursion on the original observation filtration. A sufficient primitive model has a compact state space, nonempty compact feasible sets varying continuously with state, continuous costs and invariant transitions, compact innovations with their original fixed laws, and finite monotone cash-invariant risk evaluations. These assumptions make the Bellman operator preserve continuous functions. For implemented Borel policies, the risk evaluation must also preserve Borel parameter kernels; the supplement proves this property for expectation and AVaR. Proposition 1 proves, rather than assumes, the resulting finite-horizon value existence and uniqueness, attained Borel minimizers, comparison against all feasible adapted history policies, and the dated stability bounds. These are classical foundations [Ruszczyński, 2010, 2014], not a claim of a new fixed-point principle. Pointwise transition continuity with an unbounded innovation support is insufficient for arbitrary monetary risk; the exact essential-uniform alternative is stated in the proof.

The original investment primitives satisfy these conditions directly: $F(x, a, z) \in [1/32, 27/32]^d$, $1/8 \leq A(x) \leq 1/4$, and all costs are bounded. Their joint curvature gives V_1 state strong-convexity modulus $(4 - 5\beta/4)/d$ and root action modulus two. Both dates therefore have unique continuous constrained optimal actors. Proposition 2 also derives the continuous value gradient with its capacity multiplier, the original-risk root derivative, and the actual observation/feasibility allowance. Differentiating a rounded actor is not used to establish any of these claims. The supplement records the full smooth critic–actor graph and distinguishes its gradient from a frozen-target update.

The separate Brownian recursive-utility bridge has its own existence and verification assumptions. Proposition 3 supplies the stopped-domain version, charging the original stopping boundary and terminal condition. Its Lipschitz driver can depend on the backward value, in which case cash invariance is not automatic. The continuous-time stability factor and smooth critic graph are therefore verified within that original recursive criterion.

1 Well-posed economic operators and implemented policies

This section supplies sufficient conditions and proofs for the existence, measurability, uniqueness, and stability obligations used in the paper. These are standard dynamic-programming foundations, stated explicitly rather than claimed as a new numerical contribution [Ruszczyński, 2010, 2014]. They apply directly to the original investment primitives below. A different recursive utility, information filtration, or noncompact state model requires its own corresponding hypotheses; it does not inherit these properties by being described as a Bellman problem.

1.1 A sufficient finite-horizon model

Let X and A be compact metric spaces. At each finite date $j < T$, let $A_j : X \rightrightarrows \mathsf{A}$ have nonempty compact values and be both upper and lower hemicontinuous. Its graph is denoted G_j. Let the innovation space Z_j be compact metric with fixed Borel probability P_j. The original transition $F_j : G_j \times \mathsf{Z}_j \to X$ is jointly continuous and invariant in X. The cost $c_j : G_j \to \mathbb{R}$ and terminal payoff $g : X \to \mathbb{R}$ are continuous. The finite discounts $\beta_j \geq 0$ need not be smaller than one.

Let $\rho_j : L^\infty(P_j) \to \mathbb{R}$ be finite, monotone, cash invariant, and normalized by $\rho_j(0) = 0$. Without normalization, the identical original operator is obtained using $\bar{\rho}_j = \rho_j - \rho_j(0)$ and $\bar{c}_j = c_j + \beta_j \rho_j(0)$; thus the finite deterministic constant remains in the original stage account. For implemented Borel policies we also require the following parameter-measurability property:

$$(s, z) \mapsto w(s, z) \text{ is bounded and Borel} \quad \Longrightarrow \quad s \mapsto \rho_j(w(s, \cdot)) \text{ is Borel} \tag{1}$$

for every standard Borel parameter space and bounded Borel w. This property is verified below for the original expectation and AVaR primitives; it is not deduced from cash invariance alone.

The filtration is generated by the successive original innovations, with their stipulated fixed conditional laws, and the initial state is observed. An admissible history policy is a nonanticipative Borel action rule on this history, feasible at its current original state. Its criterion is the *specified conditional nested recursion*: at a history, apply ρ_j to the next-innovation section of its continuation. This specifies the time-consistent original problem. Static AVaR of total cost and a randomization device with a different observation convention are different criteria and are not inserted into this policy class without being declared.

Proposition 1 (Finite-horizon existence, selection, and validation)**.** *Under these conditions, the original Bellman recursion has one bounded continuous value sequence $V_T = g$,*

$$V_j(x) = \min_{a \in A_j(x)} \{c_j(x, a) + \beta_j \rho_j(V_{j+1}(F_j(x, a, \cdot)))\}.$$

Each minimum is attained and admits a Borel selector. The resulting Markov policy is optimal among all admissible history policies for the stipulated recursion. Every implemented Borel feasible policy has a well-defined bounded Borel value sequence; its value is at least V_j at every date and original history with state x.

Uniqueness here concerns the value sequence. An optimizer need not be unique. If each action objective has a unique minimizer, that minimizer is continuous in x. In particular, a strongly convex objective on a compact convex feasible action set has a unique minimizer.

For the same primitives, two bounded continuous terminal continuations satisfy

$$\|V_j - \widetilde{V}_j\|_\infty \leq \left(\prod_{k=j}^{T-1} \beta_k \right) \|g - \widetilde{g}\|_\infty. \tag{2}$$

For a fixed implemented policy the same bound holds for bounded Borel terminal continuations, whose recursions are well-defined by (1)*. Additive stage-cost discrepancies of at most e_k add their dated discounted sum to the right side. Changes in a law, transition, constraints, or preference require their own operator discrepancy; they are not covered by this same-primitives statement.*

Proof. If $\|v - w\|_\infty \leq h$, then $w - h \leq v \leq w + h$. Monotonicity and cash invariance give

$$|\rho_j(v) - \rho_j(w)| \leq h, \qquad |\rho_j(v)| \leq \|v\|_\infty.$$

This also applies to equivalence classes under P_j; a stronger pointwise bound is sufficient. The feasible graph is closed in a compact product and hence compact. For continuous v, joint continuity on $G_j \times \mathsf{Z}_j$ implies uniform continuity. Consequently $(x, a) \mapsto v(F_j(x, a, \cdot))$ is continuous in the supremum norm. The displayed nonexpansiveness proves continuity of its risk evaluation, and thus of the full action objective.

Compactness of $A_j(x)$ gives an attained minimum. To prove value continuity without hiding a selection assumption, let $x_n \to x$. Along a subsequence attaining the lower limit of the values, compactness gives a convergent subsequence of minimizers $a_n \to a$. Upper hemicontinuity gives $a \in A_j(x)$, and continuity of the objective implies that the lower limit is at least $V_j(x)$. Conversely, choose a minimizer a at x. Lower hemicontinuity provides $a_n \in A_j(x_n)$ with $a_n \to a$; evaluating these feasible actions shows that the upper limit is at most $V_j(x)$. Thus the value is continuous. Backward induction from g proves existence and continuity of the complete value sequence; the finite recursion also proves its uniqueness. The norm estimate $\|V_j\|_\infty \le \|c_j\|_\infty + \beta_j \|V_{j+1}\|_\infty$ proves boundedness.

Here is an explicit measurable-selector argument. The argmin graph is closed, because both the objective and value are continuous, and its fibers are nonempty compact. For each n, choose a finite cover of A by closed balls of radius 2^{-n} with fixed centers. In a fixed ordering choose the first ball meeting the argmin fiber, and at each successive level choose the first ball meeting its intersection with the previously chosen balls. All these intersections remain nonempty compact and are nested. The event that the argmin fiber meets a fixed finite intersection of closed balls is closed: it is the projection onto compact X of a closed subset of $X \times \mathsf{A}$. Induction over the finite-valued previous choices therefore makes each new chosen center a Borel function of x. The nested sets have diameters at most 2^{1-n}, so their intersection contains exactly one action. The chosen centers converge to that action; their pointwise limit is Borel and belongs to the argmin fiber. This constructs the required selector. When the argmin is already unique, the same subsequence argument used for value continuity shows continuity of the selector.

For an arbitrary implemented history policy, composition of Borel histories, actions, and the original transition gives a bounded Borel next-history kernel. Property (1) then constructs its bounded Borel value backward from g. Suppose at the next date that the policy continuation is at least V_{j+1} at its next original state. Applying the same conditional risk at the current history preserves this inequality. The resulting current cost is at least the action objective at its feasible chosen action, hence at least V_j. Backward induction proves policy domination, without restricting the comparison to Markov policies. Using the Borel minimizing selector at every date makes every inequality an equality, which proves its optimality.

Finally, the preceding nonexpansiveness yields $\|\mathcal{T}_j v - \mathcal{T}_j w\|_\infty \le \beta_j \|v - w\|_\infty$, by comparing infima over the same actions. A fixed policy has the identical inequality without the infimum. Backward induction proves (2). When the costs differ by at most e_j, the one-step estimate has the extra term e_j; iterating it gives the dated sum. □

The exact continuity and measurability obligations. Compact innovations can be replaced by essential-uniform continuity of $v(F_j(x, a, \cdot))$ on the declared continuation class. Pointwise continuity alone is insufficient for an arbitrary monetary risk. For example, let Z have an unbounded exponential law, $F(x, z) = 1 - e^{-xz}$ for $x \in [0, 1]$, terminal value $g(y) = y$, and $\rho = \operatorname{ess\,sup}$. All these primitives are continuous and bounded where required, but $\rho(g(F(0, Z))) = 0$ and $\rho(g(F(x, Z))) = 1$ for every $x > 0$. Expectation has stronger dominated-continuity properties, but those cannot be assigned to every monotone cash-invariant risk.

For the original fixed-law AVaR and bounded jointly Borel w, the formula

$$\rho_\alpha(w(s,\cdot)) = \inf_{\eta\in\mathbb{Q}} \left\{ \eta + \frac{1}{1-\alpha} \int (w(s,z)-\eta)_+ \, P(dz) \right\}$$

proves (1). Each integral is Borel: the claim holds on measurable rectangles, extends to indicators of product-Borel sets by a monotone-class argument, and then to nonnegative Borel functions by monotone simple approximation. The infimum is countable and equals the real-threshold infimum because the displayed expression is continuous in η. Expectation is the same measurability argument without an infimum. Thus the needed property is established for both original preference cases, while remaining explicit for a different risk primitive.

1.2 The original investment domain and unique constrained actions

Proposition 2 (Original primitive closure and actor regularity)**.** *For the original even-dimensional investment primitives, every feasible state-action-innovation triple satisfies*

$$F(x,a,z) \in [1/32, 27/32]^d \subset [0,1]^d. \tag{3}$$

The capacity correspondence $[0, A(x)]$ is continuous, compact, and nonempty, with $1/8 \le A(x) \le 1/4$. Costs are continuous and nonnegative, with valid uniform bounds $S \le 49/32$, $g \le 37/16$, and $S + h \le 103/64$. Hence the finite-horizon existence and policy theorem applies for the original expectation and nested AVaR criteria.

For the two-date AVaR model with $\beta = 15/16$, V_1 is strongly convex in state with modulus $(4-5\beta/4)/d$, and the original root and last-date action objectives are strongly convex in action with modulus two. They have unique continuous constrained optimal actors. The last-date V_1 and the root action objective are continuously differentiable on the relative state interior, with the corresponding one-sided boundary interpretation.

For either date where Q_j is continuously differentiable, write $a_j^*(x)$ for its constrained minimizer and

$$\lambda_j(x) = \mathbf{1}_{\{a_j^* = A(x)\}} [-\partial_a Q_j(x, a_j^*)]_+.$$

Then the value gradient is

$$\nabla V_j(x) = \nabla_x Q_j(x, a_j^*(x)) - \lambda_j(x) \nabla A(x). \tag{4}$$

This formula is not a derivative of a rounded deployed actor. Interior actor derivatives require the additional local second derivatives and strict-interiority conditions stated in the retained compiler theorem.

Proof. For the state part of the jth transition coordinate put $p = x_j$, $q = x_{j+1}$. The function $p/2 + q/8 + p(1-q)/16$ is nondecreasing in both arguments on $[0,1]^2$, because its two partial derivatives are $1/2 + (1-q)/16$ and $1/8 - p/16$. Its range lies in $[0, 5/8]$. Since $0 \le b_j a \le 1/8$ and $|z| \le 1/32$, adding the constant $1/16$ proves (3). The affine capacity gives the stated correspondence properties. The inequalities $|x_j - 5/8| \le 5/8$, $|x_j - x_{j+1}| \le 1$, and $(1/2 - 2\bar{x})_+ \le 1/2$ give the stated state-cost bounds. Moreover $h(a) \le h(1/4) = 5/64$, which gives the full stage bound. Polynomial transitions, squared positive parts, and the fixed compact innovation interval have the required continuity. The previous measurability proof applies to the risk primitive.

For completeness, the inherited joint curvature bound follows directly from the original primitives. On each smooth region, $\nabla^2 S \succeq (4/d) I$, $\nabla^2 g \succeq 0$, and $|\partial_j g| \le 10/d$. The only nonzero second

derivative of the transition is the mixed derivative of $-x_j x_{j+1}/16$. In a joint state-action direction (v, w), the Hessian chain rule therefore gives

$$D^2(g \circ F)[(v,w)]^2 \geq -\frac{1}{8}\sum_j \partial_j g(F) v_j v_{j+1} \geq -\frac{5}{4d}\|v\|_2^2.$$

The omitted Jacobian–Hessian–Jacobian term is nonnegative in the full joint direction, including its state-action mixed terms. Together with $h'' \geq 2$, this proves for every fixed shock the lower curvature

$$\frac{4 - 5\beta/4}{d}\|v\|_2^2 + 2w^2.$$

The squared-hinge derivatives match at the junction, so integrating the almost-everywhere second-derivative inequality along a segment extends it across the junction. Integration under any feasible AVaR density preserves the bound. Subtracting its common quadratic term leaves convex functions; their supremum is convex, so the risk supremum preserves the same bound. It follows that the joint objective has the state term $(4-5\beta/4)\|x-y\|_2^2/(2d)$ in its strong-convexity inequality. Interpolating two optimal state-action pairs preserves feasibility because the capacity is affine. Taking the value at that feasible interpolation and dropping the nonnegative action-distance term proves strong convexity of V_1 with the announced modulus. This supplies the state convexity used in the active-tail theorem, rather than assuming it for a fitted continuation.

For each fixed root state, the successor is affine in its scalar action. Convexity of V_1, followed by monotonicity and convexity of AVaR, makes $a \mapsto \rho_\alpha(V_1(F(x, a, \cdot)))$ convex. Adding $h'' \geq 2$ proves root-action strong convexity; the same property of the last-date objective follows from the original primitive curvature. The preceding compact theorem now gives existence, uniqueness, and continuity of both optimal actors.

The retained exact last-date formula is C^1 in state and action: the positive-part square and its explicit tail polynomial have matching values and first derivatives at their junctions. Normalize the action by $a = A(x)v$, with $v \in [0, 1]$. The normalized objective has a unique continuous minimizer. Comparing the value increment at the old optimizer and at the new optimizer, and using uniform first-order expansion on this fixed compact action interval, gives

$$\nabla V_1 = \nabla_x Q_1 + v^* \partial_a Q_1 \nabla A.$$

At a lower endpoint $v^* = 0$, at an interior optimum $\partial_a Q_1 = 0$, and at capacity $v^* = 1$, $\partial_a Q_1 \leq 0$. This proves (4) and continuity of the value gradient. In particular the multiplier tends to zero at a switch between an interior and a binding-capacity solution.

For the root objective, invariance puts every successor strictly inside the state cube, where ∇V_1 is continuous and bounded on the compact successor range. Along the innovation, $V_1(F(x, a, z))$ is strongly convex with modulus $181/64$. Every one of its level sets has at most two points, so its uniform-law distribution has no atoms. Continuity of the function on the interval makes its distribution function strictly increasing between its minimum and maximum: the preimage of every nonempty open value interval in that range contains a nonempty open innovation interval. Its quantile is therefore unique at the declared risk levels. The differentiability conditions of the retained AVaR envelope argument apply, with bounded action/state derivatives. Equivalently, the active central interval has a unique continuous minimizer and the value envelope differentiates the integral with that interval held fixed. Thus, writing $E_{x,a}$ for the optimal two-tail set,

$$\partial_a Q_0(x, a) = h'(a) + \frac{\beta}{2R(1-\alpha)} \int_{E_{x,a}} b^\mathsf{T} \nabla V_1(F(x, a, z))\, dz.$$

The analogous state derivative has $D_x F^{\mathsf{T}} \nabla V_1$ and ∇S. Continuity of the integrands and tail endpoints gives continuity of these derivatives. Applying the same normalized action envelope proves (4) for V_0. Endpoint actions are included; no global differentiability of the actor across active-set changes is asserted. □

Observation, feasibility, and terminal-policy transfer. The implemented terminal controller observes a downward 40-bit state $y \le x$ with $\|x - y\|_1 \le d2^{-40}$ and returns an action feasible at y. Since $A(y) \le A(x)$, it remains feasible at the original state. Its certified action gap at y is at most 2^{-21}. The primitive fixed-action objective and the optimal value both have state Lipschitz constant at most $15/d$, so its original-state gap is at most

$$2^{-21} + 30\,2^{-40} < 2^{-20}.$$

The fixed-action modulus follows from $\|\nabla S\|_\infty \le 8/d$, the transition column-sum bound $11/16$, $\|\nabla g\|_\infty \le 10/d$, and $8 + \beta(11/16)10 < 15$. The value modulus is the retained capacity-aware bound. The controller's rounding and exact fallback must actually enforce its premise; an optimizer return code alone does not do so. This explains why its allowance remains valid when the new method changes the locations of original-law integration queries while retaining the same deployed controller.

1.3 Residual validation and the optional infinite-horizon case

For the finite-horizon one-sided sandwich, suppose $f_j - \mathcal{T}_j f_{j+1} \le u_j$, $\mathcal{T}_j^\pi f_{j+1} - f_j \le v_j$, and $\ell_T \le f_T - g \le u_T$. Monotonicity and cash invariance give separately

$$f_0 - V_0 \le \sum_{j<T} B_j u_j + B_T u_T, \qquad J_0^\pi - f_0 \le \sum_{j<T} B_j v_j - B_T \ell_T.$$

Each follows by backward induction, starting from its corresponding terminal inequality. Adding them proves the retained complete-policy bound, and feasibility supplies $J_0^\pi \ge V_0$. This validates an NBO's own attained continuation without assuming optimal-continuation training labels. Date-specific constant shifts of f_j cancel in the paired residuals and in $u_T - \ell_T$.

For a stationary infinite-horizon model with the same compact and kernel conditions, bounded cost, and $0 \le \beta < 1$, the Bellman map is a β-contraction on the complete space $C(X)$. The finite-horizon continuity proof shows that it maps this space into itself. Thus it has a unique continuous fixed point V. For any Borel feasible stationary policy, the policy map is a β-contraction on bounded Borel functions by (1); it has a unique bounded Borel fixed point J^π. A Borel minimizer of the continuous action objective at V satisfies $T_{\pi^*} V = TV = V$, so uniqueness of its policy fixed point gives $J^{\pi^*} = V$. For another policy, $V \le T_\pi V$; monotone iteration and contraction give $V \le J^\pi$.

The infinite-history criterion is well-defined as the limit of the same finite nested recursions: the omitted cost after n dates is bounded uniformly by $\|c\|_\infty \beta^n/(1-\beta)$. Passing the finite-horizon comparison to this limit proves optimality against the same admissible history-policy class. This argument does not assert that infimizing arbitrary Borel action objectives preserves Borel measurability. The more general policy-iteration theorem in the retained supplement explicitly assumes that additional preservation and selection property when it applies a Bellman infimum to possibly discontinuous policy values.

1.4 The critic and actor computational graph

For a smooth parameterized critic and differentiable feasible actor, the derivative must correspond to the objective actually optimized. For example, at a fixed original state write

$$r_\theta = c(x, a_\theta) + \beta\rho(f_{+,\theta}(F(x, a_\theta, Z))) - f_\theta(x).$$

Suppose differentiation is justified by integrable local domination, and either ρ is expectation or it is AVaR at an atomless unique quantile, as in the retained tail-gradient theorem. Let q_θ be its maximizing tail density, with $q_\theta = 1$ for expectation. The chain and risk-envelope rules give

$$\begin{aligned}\nabla_\theta r_\theta = (D_\theta a_\theta)^\mathsf{T} \left[\nabla_a c + \beta\mathbb{E}_{q_\theta}(D_a F)^\mathsf{T}\nabla_x f_{+,\theta}(F)\right] \\ + \beta\mathbb{E}_{q_\theta}\nabla_\theta f_{+,\theta}(F) - \nabla_\theta f_\theta(x).\end{aligned}$$

The last expectation is a partial parameter derivative at a fixed successor. The actor contribution is written separately to avoid double counting. A risk-envelope derivative holds the maximizing density fixed in this formula; its validity comes from the stated regularity assumptions, not from differentiating a sorting permutation. The half-squared-residual objective $\mathcal{L}(\theta) = \frac{1}{2}\mathbb{E}[r_\theta^2]$ has gradient $\mathbb{E}[r_\theta\nabla_\theta r_\theta]$ when its own domination condition holds. Detaching the future target defines a different frozen-target gradient or a semigradient algorithm. It does not establish that this full original coupled gradient was used, and neither formula proves optimizer convergence.

At a projection or rounding discontinuity these differentiability premises may fail. Piecewise training derivatives can be specified as an algorithmic choice, but the global economic certificate still uses the actual feasible action and its original-law inequalities. This is consistent with the own-future constructive theorem, whose validity does not depend on a successful differentiable optimizer.

2 Stopping boundaries and the continuous-time recursion

The retained Brownian Markov bridge is a separate classical recursive utility verification theorem [Pardoux and Peng, 1990, El Karoui et al., 1997]. Its forward coefficients and driver are understood to be jointly Borel in their displayed deterministic arguments, so composition with each admissible progressively measurable control yields progressively measurable coefficients. The forward state-Lipschitz constants and driver (y, z)-Lipschitz constants are uniform in the control. Terminal payoff and the driver at $(y, z) = (0, 0)$ are bounded. A feedback must define its own well-posed closed-loop state process. These conditions make the retained existence proof precise; they do not establish existence for a fully coupled FBSDE or an arbitrary discontinuous feedback.

Proposition 3 (Stopped original-BSDE verification)**.** *Let $\tau^a \le T$ be a specified stopping time for each admissible control, such as exit from a declared domain capped at T, with bounded Borel original stopping payoff $G(\tau^a, X_{\tau^a})$. Before stopping use exactly the driver and forward coefficients of the retained BSDE bridge. Let bounded $u \in C^{1,2}$ have the bounded derivatives required for stopped Itô calculus and continuous values on the declared stopping boundary. Suppose*

$$\mathcal{H}^a u \ge -e_-(s) \text{ for every admissible action,} \qquad \mathcal{H}^\pi u \le e_+(s)$$

on the unstopped domain, and at every possible stopping location and at the terminal part of the boundary suppose

$$-\eta_- \le G(s, x) - u(s, x) \le \eta_+.$$

Here $e_\pm$ are nonnegative deterministic integrable functions and $\eta_\pm \geq 0$. The stopped BSDE has a unique solution in $S^2 \times H^2$, and

$$0 \leq Y_0^\pi - \inf_a Y_0^a \leq e^{L_y T}(\eta_- + \eta_+) + \int_0^T e^{L_y s}(e_-(s) + e_+(s))\, ds.$$

The stopping rule and boundary payoff are parts of the original economic criterion. A smaller interior residual does not remove their errors or permit them to be changed.

Proof. Embed the stopped equation on $[0,T]$ with terminal variable $\xi^a = G(\tau^a, X_{\tau^a})$ and driver $\mathbf{1}_{\{s<\tau^a\}} f(s, X_s, y, z, a_s)$. It is progressively measurable, has the same uniform Lipschitz constants, and has square-integrable terminal data and bounded driver at zero. The weighted Picard argument in the retained proof therefore gives a unique $S^2 \times H^2$ solution. After τ^a it equals ξ^a with $Z = 0$; before stopping it is exactly the declared stopped equation.

Apply stopped Itô calculus to $u(s \wedge \tau^a, X_{s\wedge\tau^a})$. Subtracting gives the same linearized error BSDE as in the retained proof, with its coefficients set to zero after τ^a, terminal error $\xi^a - u(\tau^a, X_{\tau^a})$, and residual driver $\mathbf{1}_{\{s<\tau^a\}} \mathcal{H}^{a_s} u$. Its measurable coefficients satisfy $|A_s| \leq L_y$ and $\|\vartheta_s\| \leq L_z$. The bounded Girsanov coefficient has an exponential density with all finite moments. The same Cauchy–Schwarz and H^1 martingale argument used in the retained verification proof therefore justifies expectation of the stopped integration-factor identity,

$$Y_0^a - u(0,x) = \mathbb{E}^\vartheta \left[e^{\int_0^{\tau^a} A_v dv} (G - u)(\tau^a, X_{\tau^a}) + \int_0^{\tau^a} e^{\int_0^s A_v dv} \mathcal{H}^{a_s} u(s, X_s)\, ds \right].$$

The factor is positive and at most $e^{L_y s}$. Applying the lower residual and boundary bounds gives a lower bound for every admissible control. Applying the upper bounds gives an upper bound for the implemented policy. Since $\tau^a \leq T$, the deterministic integrals and terminal factors are bounded by those displayed in the proposition. Taking the infimum of the lower bounds and subtracting proves the result. □

Why the boundary and graph are material. For a deterministic state starting at $1/2$ in $(0,1)$, use either of the two constant controls $a = \pm 1$ in $dX_s = a\, ds$, stop on exit before $T = 1$, and assign boundary costs zero at the left endpoint and one at the right. With zero driver and $u = 0$, every interior residual is zero, but the right-exit policy has regret one. The stopping-boundary error exactly supplies the missing term. This is a verification counterexample, not a second numerical application.

The BSDE graph is $z_u = \sigma^\mathsf{T} \nabla u$ for the same control and the original driver. Its $C^{1,2}$ requirement is substantive. A piecewise-affine ReLU critic is generally not C^2 and cannot be inserted into this classical formula by declaring its second derivative zero away from its kinks; a different weak, viscosity, or smoothing argument with the required error account would first be needed. The finite-horizon ReLU Bellman construction uses order and Lipschitz arguments and does not rely on this continuous-time smooth verification theorem.

Finally, a general recursive utility is not automatically cash invariant. Even the globally Lipschitz BSDE driver $f(s,y,z) = \lambda y$ gives $Y_0 = e^{\lambda T}\mathbb{E}\xi$, so adding a constant c to the terminal criterion changes Y_0 by $e^{\lambda T}c$, not generally by c. The continuous-time bridge correctly uses its $e^{L_y T}$ stability factors. It does not inherit the monetary discrete-risk contraction or identify an arbitrary NDU/FBSDE with the original investment AVaR objective.

References

Nicole El Karoui, Shige Peng, and Marie-Claire Quenez. Backward stochastic differential equations in finance. *Mathematical Finance*, 7(1):1–71, 1997. doi: 10.1111/1467-9965.00022.

Etienne Pardoux and Shige Peng. Adapted solution of a backward stochastic differential equation. *Systems & Control Letters*, 14(1):55–61, 1990. doi: 10.1016/0167-6911(90)90082-6.

Andrzej Ruszczyński. Risk-averse dynamic programming for Markov decision processes. *Mathematical Programming*, 125(2):235–261, 2010. doi: 10.1007/s10107-010-0393-3.

Andrzej Ruszczyński. Erratum to: Risk-averse dynamic programming for Markov decision processes. *Mathematical Programming*, 145:601–604, 2014. doi: 10.1007/s10107-014-0783-z.